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Techno-Economic Planning of Decentralized Net-Zero Energy Systems for Low-Carbon Development in Uganda

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Abstract: Uganda's growing electricity demand, persistent rural electrification gaps, and dependence on hydropower create an urgent need for diversified and decentralized low-carbon energy systems. However, the economic viability and optimal technology composition of such systems vary across locations and electricity-use settings. This study investigates the cost-optimal design of grid-connected and off-grid net-zero energy systems under representative Ugandan climatic, economic, and policy conditions. A mixed-integer linear programming (MILP) framework was developed to evaluate solar photovoltaic systems, battery energy storage, biomass generators, and small-scale wind turbines for residential, institutional, and agro-processing applications. Three representative locations were examined: Kampala for a predominantly grid-connected urban and institutional setting, Gulu District for rural and off-grid electrification, and Mbarara District for agro-processing and productive electricity use. The model minimized total lifecycle cost subject to annual net-zero energy balance, reliability requirements, and a minimum greenhouse gas emission reduction of 80% relative to diesel-based generation. The results showed that solar photovoltaic and battery systems formed the principal cost-optimal configurations, with levelized cost of energy (LCOE) ranging from USD 0.072 to 0.118/kWh. Biomass generation was selected for institutional and agro-processing applications, where it reduced storage requirements and supported electricity supply during periods of extended demand. The optimized systems reduced greenhouse gas emissions by approximately 89–91% relative to diesel-based alternatives. Sensitivity analysis showed that battery capital cost and financing conditions had the greatest influence on system economics. A 20% reduction in battery cost lowered the LCOE by approximately 11–16%, whereas an increase in the real discount rate from 8% to 12% raised it by approximately 9–14%. These findings demonstrate that decentralized net-zero energy systems can support electrification and low-carbon development under the representative Ugandan conditions examined. The study provides a planning framework for technology screening, financing assessment, and comparison of low-carbon electricity strategies, while confirming the need for site-specific resource, demand, infrastructure, and financial evaluation before project implementation.

Keywords: Decentralized net-zero energy systems; Techno-economic optimization; Low-carbon development; Solar photovoltaic energy; Battery energy storage; Biomass generation; Greenhouse gas mitigation; Uganda

1 Introduction

Uganda's electricity sector is undergoing rapid transformation as population growth, urbanization, expanding social services, and productive economic activities increase the demand for reliable electricity. Although the country's installed generation capacity has exceeded 1,300 MW, electricity access remains uneven [1–3]. The disparity is particularly evident in rural and peri-urban areas, where low population density, dispersed demand, and infrastructure constraints make grid extension technically difficult and economically costly. Reliability concerns also persist in grid-connected areas because Uganda's electricity supply remains strongly dependent on large hydropower plants that are exposed to hydrological variability [3]. Diesel generators consequently continue to meet part of the unmet or backup electricity demand of households, health facilities, and agro-processing enterprises. Their continued use raises electricity costs and produces substantial greenhouse gas emissions [4–6]. Addressing these interconnected

problems requires electricity-planning strategies that combine wider access, dependable supply, financial viability, and long-term decarbonization.

Decentralized net-zero energy systems offer one possible route towards these objectives. Such systems are designed to generate, on an annual basis, an amount of energy equivalent to their total consumption. They can combine renewable generation, energy storage, grid exchange, and, where appropriate, dispatchable low-carbon resources to meet electricity demand while limiting reliance on fossil-fuel generation. Uganda has favorable conditions for this form of decentralized energy development. Average solar irradiance typically ranges from 5.1 to 5.8 kWh/m^2/day, while agricultural production generates residues that may be used for biomass-based electricity generation [6–8]. At the same time, declining costs of solar photovoltaic modules and lithium-ion battery storage are improving the economic prospects of renewable electricity systems in both grid-connected and off-grid settings [9]. These conditions create an opportunity to connect electrification planning with Uganda's transition towards a lower-carbon economy.

Resource availability alone, however, does not establish whether a decentralized net-zero system is technically reliable or economically justified. Its performance depends on the combination and size of generation technologies, the relationship between hourly supply and demand, the amount of storage required, the availability of dispatchable generation, and the financing conditions applied over the project lifetime. Energy planners and investors therefore require quantitative evidence on cost-optimal system configurations and on the trade-offs between battery storage and dispatchable renewable generation. They must also determine whether a system can maintain an acceptable level of reliability while satisfying annual net-zero and greenhouse gas reduction requirements. Electrification strategies that focus on a single technology or minimize generation cost without explicitly representing these requirements may produce systems that are oversized, financially inefficient, or inconsistent with long-term low-carbon development objectives.

Previous studies have examined the technical and economic feasibility of hybrid renewable energy systems in Sub-Saharan Africa. Much of this work has concentrated on solar photovoltaic (PV)–battery and PV–diesel systems for rural electrification [10–12]. Previous research has investigated low-carbon energy transition pathways across Africa [13], while other studies have examined the potential of biomass resources for electricity generation and clean energy development in Uganda [14, 15]. These studies establish the relevance of hybrid renewable systems to the region, but several limitations remain. Many rely on simulation-based or heuristic sizing procedures, while environmental performance is commonly calculated after the system has been designed rather than imposed as a condition of the optimization. In addition, most studies examine a single location, demand category, or technology configuration. Consequently, their results do not clearly show how different load characteristics influence cost-optimal technology selection across residential, institutional, and productive-use applications.

Optimization-based energy-system models provide a structured means of addressing these limitations. Mixed-integer linear programming (MILP) can determine technology capacities and operational schedules simultaneously while representing technical, economic, reliability, and environmental constraints within a common mathematical formulation [16–18]. MILP methods have been widely used in power-system planning and microgrid design, particularly in high-income settings. Their application to decentralized net-zero energy planning in low- and middle-income countries remains less developed, especially where annual net-zero balance and minimum emission-reduction requirements are imposed directly within the optimization. For Uganda, existing research has not systematically compared net-zero configurations across multiple end-use settings while accounting for regional resource conditions, hourly electricity demand, reliability requirements, lifecycle costs, and financing conditions within the same analytical framework.

This study addresses these gaps by developing and applying a MILP-based techno-economic planning framework for decentralized net-zero energy systems under representative Ugandan climatic, economic, and policy conditions. Grid-connected and off-grid configurations were evaluated for residential, institutional, and agro-processing applications. The framework jointly considered solar photovoltaic generation, battery energy storage, biomass generation, small-scale wind power, and grid interaction where applicable. It minimized total lifecycle cost while enforcing hourly energy balance, annual net-zero performance, an acceptable reliability threshold, and a minimum greenhouse gas emission reduction relative to diesel-based generation. The analysis also examined how battery capital cost, financing conditions, and biomass fuel price affected the economic performance and stability of the selected configurations. By treating emission reduction as an explicit design constraint, rather than solely as an indicator calculated after optimization, the framework tested whether economic and low-carbon objectives could be satisfied simultaneously.

The study makes three principal contributions. First, it provides a consistent comparison of cost-optimal net-zero energy systems across household, health-center, and agro-processing demand profiles representing contrasting Ugandan electricity-use conditions. Second, it clarifies how solar PV, battery storage, and biomass generation interact under annual net-zero, reliability, and emission-reduction constraints. Third, it identifies the technology-cost and financing parameters that exert the greatest influence on the levelized cost of decentralized electricity.

In this way, the study connects energy-system optimization with the economic and environmental requirements of low-carbon development. Its findings provide a quantitative basis for technology screening, scenario comparison, and preliminary energy planning, while recognizing that project-level investment decisions require site-specific assessment of electricity demand, renewable resources, infrastructure, biomass supply, and financing conditions.

2 Methodology

This study adopts a techno-economic optimization framework based on MILP to design and evaluate net-zero energy systems tailored to Uganda's climatic, economic, and infrastructural conditions. The methodological approach integrates region-specific resource assessment, load modeling, component-level performance representation, lifecycle cost analysis, environmental constraints, and system optimization within a unified mathematical formulation. The framework is designed to identify cost-optimal system configurations that satisfy annual net-zero energy balance, reliability, and emission reduction targets over the full project lifetime.

2.1 Study Areas and Spatial Representation

The analysis is conducted for three representative locations in Uganda selected to capture differences in electricity demand, grid accessibility, renewable resource availability, and productive energy use. The selected locations are Kampala in the Central Region, Gulu District in the Northern Region, and Mbarara District in the Western Region. These locations are used as representative modeling cases rather than as an exhaustive spatial characterization of the entire country. Kampala is used to represent the Central Region and the predominantly grid-connected case. As Uganda's principal urban and commercial center, Kampala is characterized by comparatively high electricity demand, extensive grid availability, dense residential and institutional loads, and significant reliance on the interconnected electricity system. The Kampala case therefore provides an appropriate representation of conditions in which distributed renewable generation and battery storage operate in conjunction with grid imports and exports.

Gulu District is used to represent the Northern Region and the predominantly off-grid or weak-grid rural electrification case. Although the regional urban center has grid access, many surrounding communities in northern Uganda experience lower electrification levels, dispersed settlement patterns, and higher costs of conventional grid extension. The Gulu case is therefore modeled to represent rural household and essential-service applications for which autonomous PV–battery or hybrid renewable systems can provide an alternative to diesel generation or unreliable grid supply.

Mbarara District is used to represent the Western Region and the agro-processing case. Western Uganda has substantial agricultural production and associated productive-use electricity requirements, including milling, cooling, processing, pumping, and other activities with relatively high and sustained demand. Agricultural residues generated within such production systems also create opportunities for locally sourced biomass energy. Mbarara is therefore used as a representative setting for evaluating PV–battery–biomass systems in agro-processing applications.

For each representative location, long-term solar resource conditions are represented using typical meteorological year data. Biomass availability in the Western case is estimated from agricultural residue availability using conservative residue-to-product ratios and availability factors to avoid assuming that all residues can be diverted to electricity production. The three locations are intended to capture contrasting system-planning conditions and should therefore be interpreted as representative cases rather than as site-specific predictions for all locations within their respective regions.

2.2 System Boundary, Temporal Resolution, and Load Modeling

The net-zero energy systems are modeled over a project lifetime of twenty years using an hourly temporal resolution, resulting in 8,760 operational time steps for the representative year. The system boundary includes electricity generation, battery storage, electricity consumption, biomass-fuel use where applicable, and grid interaction for the grid-connected case. For grid-connected systems, the grid is represented as both an electricity source and sink under a hypothetical annual net-metering assumption, subject to regulatory verification, whereas off-grid configurations are required to meet electricity demand autonomously subject to the specified reliability constraint. Three hypothetical hourly load profiles are developed to represent a rural household, a health center, and an agro-processing plant. Although these profiles are not based on direct measurements from individual facilities, they are structured to reproduce realistic differences in daily energy consumption, peak demand, operating duration, and temporal demand patterns. The profiles are therefore intended for comparative system optimization rather than prediction of electricity use at a particular site.

The rural household profile is dominated by lighting, phone charging, refrigeration, television, small appliances, and limited productive-use equipment, producing pronounced morning and evening peaks. The health-center profile includes continuous essential demand from refrigeration, communications, security lighting, laboratory equipment, and selected medical loads, together with higher daytime consumption associated with clinical activity. The agro-processing profile represents substantially larger productive-use demand associated with motors, processing

machinery, pumping, material handling, cooling, and auxiliary equipment. Seasonal and hourly variations are incorporated into the profiles while preserving the representative daily consumption and peak-load characteristics shown in Table 1. These profiles are held constant for the baseline optimization so that differences between scenarios primarily reflect resource availability, technology configuration, and economic parameters rather than uncontrolled differences in demand assumptions.

Table 1. Characteristics of representative electricity load profiles

Demand Case	Representative Location/Context	Average Daily Electricity Demand	Peak Load	Typical Operating Pattern
Rural household	Gulu District/off-grid	11.5 kWh/day	2.2 kW	Main demand from 05:30–08:00 and 18:00–23:00, with low daytime base load.
Health center	Kampala/Central institutional case	82 kWh/day	11.5 kW	Essential loads operate 24 h/day, with higher activity from approximately 07:00–20:00.
Agro-processing plant	Mbarara District/productive-use case	460 kWh/day	72 kW	Production load concentrated approximately 07:00–18:00, with residual refrigeration, storage, and auxiliary loads outside processing hours.

2.3 Objective Function and Economic Formulation

The optimization objective is to minimize the total net present cost (NPC) of the system over the project lifetime, as defined in Eq. (1), which aggregates capital investment, replacement, operation and maintenance, fuel, and salvage value components discounted to present value. Capital costs for solar photovoltaic systems, lithium-ion battery storage, and biomass gasification units are assumed to be USD 750/kW, USD 350/kWh, and USD 1,200/kW, respectively, reflecting prevailing market conditions in East Africa. A real discount rate of eight percent is applied to represent typical financing conditions for energy infrastructure projects in Uganda, and diesel fuel cost for the baseline comparison is set at USD 1.25 per liter. Replacement and operation and maintenance costs are treated as recurring expenses and discounted annually, while salvage value is credited at the end of the project lifetime based on remaining component life.

2.4 Energy Balance and Net-Zero Constraints

System operation is governed by an hourly energy balance constraint, given in Eq. (7), which ensures that electricity supply from renewable generation, storage discharge, and grid imports equals electricity demand, storage charging, and grid exports at every time step. Net-zero energy performance is enforced on an annual basis. For grid-connected systems, this is achieved by equating total annual grid imports and exports, as expressed in Eq. (8), thereby ensuring zero net electricity consumption from the grid. For off-grid systems, internal net-zero balance is enforced by equating total annual renewable electricity generation to total annual load demand, as shown in Eq. (9).

2.5 Renewable Generation and Storage Modeling

Solar photovoltaic generation is modeled using a linear performance relationship between installed capacity and plane-of-array solar irradiance, as defined in Eq. (10), assuming fixed-tilt systems oriented at regionally appropriate angles. Lithium-ion battery energy storage systems are represented using state-of-charge (SOC) dynamics that capture charging and discharging efficiencies, operational limits, and hourly energy balance, as described by Eqs. (13) through (16). Battery degradation was assumed at 2% annually over the 20-year project lifetime and incorporated into the SOC constraints through annual reductions in usable storage capacity. Battery replacement was scheduled after 12 years, when capacity falls below 80% of its initial value, with replacement costs discounted at 8% and included in the NPC. Biomass gasification systems are modeled as dispatchable generators subject to capacity constraints, with fuel consumption linked directly to electrical output through conversion efficiency and fuel lower heating value, as given in Eqs. (11) and (12). Small-scale wind turbines are included as optional components in the optimization; however, their contribution is constrained by low average wind speeds in most regions and is therefore endogenously selected only when economically justified.

2.6 Reliability and Environmental Constraints

System reliability is quantified using the Loss of Load Probability (LLP), defined in Eq. (17), which represents the fraction of total electricity demand that is unmet over the simulation period. To ensure an acceptable level of

service, the optimization imposes a maximum LLP of 1%, as specified in Eq. (18). Environmental performance is incorporated through an emissions-accounting framework in which the optimized systems are compared with a diesel-only reference case. The model imposes a minimum greenhouse gas emission reduction requirement of 80% relative to the diesel baseline. The principal emissions considered are those associated with diesel generation, grid electricity consumption where applicable, and the non-combustion biomass supply chain in biomass-integrated configurations.

Biogenic carbon dioxide released during combustion of sustainably sourced agricultural residues is treated as carbon-neutral at the point of generation because the carbon released during conversion is assumed to have been previously absorbed during biomass growth. This assumption does not imply that biomass electricity has zero lifecycle emissions. Upstream activities associated with residue collection, aggregation, transportation, drying, handling, and preprocessing require energy and can generate fossil greenhouse gas emissions. Accordingly, the biomass-integrated case includes a simplified upstream biomass supply-chain emission allowance while excluding biogenic combustion CO_2 from the net emission total.

The emissions system boundary should therefore be interpreted as a simplified lifecycle-equivalent boundary rather than a full consequential lifecycle assessment. It accounts for the dominant modeled electricity- and fuel-related emission sources but does not include detailed emissions associated with equipment manufacture, construction, land-use change, infrastructure development, or end-of-life treatment. Eq. (19) is applied consistently with this system boundary, while Eq. (20) requires the optimized configuration to achieve at least an 80% reduction relative to the diesel-only reference case.

2.7 Economic Performance Indicators and Sensitivity Analysis

Beyond the optimization objective, system performance is evaluated using standard economic indicators, including the levelized cost of energy (LCOE), defined in Eq. (21), and discounted payback period, defined in Eq. (22). Sensitivity analysis is conducted to assess the robustness of the optimized configurations to changes in parameters that are uncertain or expected to vary over the project lifetime. Battery capital cost is varied by ±20% relative to the baseline value of USD 350/kWh. The real discount rate is varied around the baseline value of 8%, with representative low- and high-financing-cost cases of 6% and 12%, respectively. Biomass fuel price is varied by ±15% in the biomass-inclusive scenarios. For each variation, the model is re-evaluated while the remaining assumptions are held at their baseline values. The sensitivity analysis focuses primarily on changes in LCOE and the stability of technology selection because these indicators directly reveal whether the principal conclusions depend strongly on assumed technology costs or financing conditions. The analysis therefore provides a quantitative basis for identifying the economic parameters most important to the feasibility of decentralized net-zero energy systems.

2.8 Assumptions and Limitations

The modeling framework is developed under a set of simplifying assumptions to ensure tractability and transparency. Solar irradiance conditions are represented using deterministic typical meteorological data and are assumed to remain broadly representative over the project lifetime. Battery degradation is represented using a constant annual rate, biomass feedstock availability is assumed to remain adequate within the specified resource constraint, and the policy and regulatory environment is assumed to remain unchanged over the modeled period.

The analysis also uses representative hypothetical load profiles rather than measured electricity-demand data from individual households, health centers, or agro-processing facilities. Consequently, site-specific behavioral variations, demand growth, equipment start-up characteristics, unusual operating schedules, and short-duration demand spikes may not be fully represented. Similarly, deterministic weather inputs do not capture the complete probability distribution of extended cloudy periods, extreme events, or interannual resource variability.

A further limitation concerns biomass emissions accounting. Although biogenic CO_2 from sustainably sourced agricultural residues is treated as carbon-neutral at the point of electricity generation, biomass supply is not completely emission-free. The present analysis includes only a simplified allowance for upstream residue collection, transport, handling, and preprocessing and does not constitute a full biomass lifecycle assessment. Site-specific transport distances, preprocessing technologies, competing residue uses, soil-carbon implications, and possible indirect land-use effects could alter the environmental performance of biomass-based systems.

The three geographical cases are also representative rather than exhaustive. Kampala, Gulu District, and Mbarara District are used to characterize contrasting grid-connected, rural/off-grid, and agro-processing conditions, respectively, but they cannot capture the complete range of resource, infrastructure, socioeconomic, and demand conditions found throughout Uganda. Accordingly, the results should be interpreted as comparative planning and scenario-analysis outcomes rather than precise predictions for individual projects. Direct investment decisions would require site-specific resource measurements, measured or audited load data, detailed infrastructure assessment, current tariffs and financing conditions, local biomass supply-chain evaluation, and stochastic analysis of resource and demand uncertainty.

3 Mathematical Formulation

The techno-economic optimization of the net-zero energy systems is formulated as a MILP problem that simultaneously determines optimal technology capacities and hourly operational decisions. The model minimizes the total lifecycle cost of the system while satisfying energy balance, net-zero energy, reliability, and emission reduction constraints over a predefined project lifetime.

3.1 Objective Function

The optimization objective is to minimize the total NPC of the system over the project lifetime [19]:

$$\min \mathrm{NPC} = C_{\mathrm{cap}} + C_{\mathrm{rep}} + C_{\mathrm{O\&M}} + C_{\mathrm{fuel}} - C_{\mathrm{salv}} \tag{1}$$

Capital investment cost is expressed as:

$$C_{\mathrm{cap}} = \sum_i C_i^{\mathrm{cap}} X_i \tag{2}$$

where, X_i represents the installed capacity of component i. Replacement costs over the project lifetime N are discounted as:

$$C_{\mathrm{rep}} = \sum_i \sum_{y=1}^{N} \frac{C_{i,y}^{\mathrm{rep}}}{(1+r)^y} \tag{3}$$

Annual operation and maintenance costs are calculated as:

$$C_{\mathrm{O\&M}} = \sum_i \sum_{y=1}^{N} \frac{C_{i,y}^{\mathrm{O\&M}}}{(1+r)^y} \tag{4}$$

Fuel costs associated with dispatchable generators are given by:

$$C_{\mathrm{fuel}} = \sum_{t=1}^{T} \frac{C_f F_t}{(1+r)^{y(t)}} \tag{5}$$

The salvage value of system components at the end of the project lifetime is defined as:

$$C_{\mathrm{salv}} = \sum_i \frac{C_i^{\mathrm{salv}}}{(1+r)^N} \tag{6}$$

3.2 Energy Balance Constraint

At each hourly time step, total electricity supply must equal served electricity demand, battery charging, grid exports, and curtailed renewable generation. Unmet electricity demand is explicitly represented to accommodate the maximum allowable LLP of 1%, while battery conversion losses are accounted for through the storage SOC dynamics [20]:

$$P_t^{\mathrm{PV}} + P_t^{\mathrm{Bio}} + P_t^{\mathrm{Wind}} + P_t^{\mathrm{Bat,dis}} + P_t^{\mathrm{Grid,imp}} = L_t - P_t^{\mathrm{unmet}} + P_t^{\mathrm{Bat,ch}} + P_t^{\mathrm{Grid,exp}} + P_t^{\mathrm{curt}} \tag{7}$$

3.3 Net-Zero Energy Constraint

For grid-connected systems, annual net-zero energy balance is enforced by equating total grid imports and exports:

$$\sum_{t=1}^{T} P_t^{\mathrm{Grid,imp}} = \sum_{t=1}^{T} P_t^{\mathrm{Grid,exp}} \tag{8}$$

For off-grid systems, annual net-zero energy balance is enforced using Eq. (9), which accounts for renewable electricity generation, electricity demand served, battery conversion losses, renewable energy curtailment, and other modeled system losses. A cyclic battery SOC condition is imposed to ensure that the annual energy balance is satisfied without net depletion of stored energy.

$$\begin{aligned}\sum_{t=1}^{T} \left(P_t^{\mathrm{PV}} + P_t^{\mathrm{Bio}} + P_t^{\mathrm{Wind}}\right) \Delta t = \sum_{t=1}^{T} \left(L_t - P_t^{\mathrm{unmet}}\right) \Delta t \\ + \sum_{t=1}^{T} P_t^{\mathrm{curt}} \Delta t + \sum_{t=1}^{T} P_t^{\mathrm{loss}} \Delta t + \Delta \mathrm{SOC}\end{aligned} \tag{9}$$

3.4 Renewable Energy Generation Models

Solar photovoltaic output is modeled as a linear function of installed capacity and plane-of-array irradiance [21]:

$$P_t^{\mathrm{PV}} = X_{\mathrm{PV}} \eta_{\mathrm{PV}} \frac{G_t}{G_{\mathrm{STC}}} \tag{10}$$

Biomass-based electricity generation is treated as dispatchable and constrained by installed capacity [22]:

$$0 \leq P_t^{\mathrm{Bio}} \leq X_{\mathrm{Bio}} \tag{11}$$

Biomass fuel consumption is linked to electrical output through conversion efficiency [16]:

$$F_t^{\mathrm{Bio}} = \frac{P_t^{\mathrm{Bio}}}{\eta_{\mathrm{Bio}} \cdot \mathrm{LHV}_{\mathrm{Bio}}} \tag{12}$$

3.5 Battery Energy Storage Model

Battery SOC evolution is governed by [23]:

$$\mathrm{SOC}_t = \mathrm{SOC}_{t-1} + \eta_{\mathrm{ch}} P_t^{\mathrm{Bat,ch}} \Delta t - \frac{1}{\eta_{\mathrm{dis}}} P_t^{\mathrm{Bat,dis}} \Delta t \tag{13}$$

Operational limits are imposed as:

$$\mathrm{SOC}_{\min} \leq \mathrm{SOC}_t \leq \mathrm{SOC}_{\max} \tag{14}$$

$$0 \leq P_t^{\mathrm{Bat,ch}} \leq X_{\mathrm{Bat}} \tag{15}$$

$$0 \leq P_t^{\mathrm{Bat,dis}} \leq X_{\mathrm{Bat}} \tag{16}$$

3.6 Reliability Constraint

System reliability is quantified using the LLP [24]:

$$\mathrm{LLP} = \frac{\sum_{t=1}^{T} E_t^{\mathrm{unserved}}}{\sum_{t=1}^{T} L_t} \tag{17}$$

The model enforces a maximum allowable LLP of 1%:

$$\mathrm{LLP} \leq 0.01 \tag{18}$$

3.7 Emissions Constraint

Annual greenhouse gas emissions are calculated by combining the relevant emission-producing activities within the defined system boundary. These include grid electricity imports, diesel consumption in the reference system, and upstream biomass supply-chain activities where biomass generation is used. Biogenic carbon dioxide released during combustion of sustainably sourced agricultural residues is excluded from the net total under the point-of-use carbon-neutrality assumption. Conceptually, annual system emissions can be expressed as the sum of grid-related emissions, diesel-related emissions where applicable, and upstream biomass supply-chain emissions:

$$E_{\mathrm{system}} = E_{\mathrm{grid}} + E_{\mathrm{diesel}} + E_{\mathrm{biomass\ supply}} \tag{19}$$

where, the biomass term represents non-biogenic emissions associated with collection, transportation, handling, and preprocessing rather than the CO_2 released during biomass conversion itself. Emission reduction relative to the diesel-only baseline is calculated as [25]:

$$\text{Emission reduction } (\%) = \frac{E_{\mathrm{diesel,baseline}} - E_{\mathrm{system}}}{E_{\mathrm{diesel,baseline}}} \times 100 \tag{20}$$

The optimization requires the resulting emission reduction to be at least 80%.

3.8 Economic Performance Metrics

The LCOE is computed as [26]:

$$\mathrm{LCOE} = \frac{\sum_{y=1}^{N} \frac{C_y}{(1+r)^y}}{\sum_{y=1}^{N} \frac{E_y}{(1+r)^y}} \tag{21}$$

For a specified avoided-cost or revenue baseline, the discounted payback period may be defined as [27]:

$$\mathrm{Payback} = \min \left\{ y \;\middle|\; \sum_{k=1}^{y} \frac{B_k - C_k}{(1+r)^k} \geq 0 \right\} \tag{22}$$

4 Results

The results presented in this section correspond to the optimal solutions of the MILP problem formulated in Section 3. All reported configurations satisfy the cost minimization objective in Eq. (1), the hourly energy balance constraint in Eq. (7), the annual net-zero conditions in Eqs. (8)–(9), the reliability requirement in Eqs. (17)–(18), and the emission reduction constraint in Eq. (20). Unless otherwise stated, all scenarios were feasible under the imposed constraints.

4.1 Optimal System Configurations

Table 2 presents the optimal installed capacities and corresponding LCOE, computed using Eq. (21), for the three representative demand cases.

Table 2. Optimal net-zero energy system configurations and economic performance

Scenario	PV Capacity (kW)	Battery Capacity (kWh)	Biomass Capacity (kW)	LCOE (USD/kWh)
Rural household	5	10	0	0.118
Health center	25	60	10	0.094
Agro-processing plant	120	200	60	0.072

Note: PV = photovoltaic; LCOE = levelized cost of energy.

Solar photovoltaic capacity is selected in all scenarios as the dominant generation technology due to its low marginal cost and favorable irradiance conditions embedded in Eq. (10). The rural household case satisfies the energy balance constraint in Eq. (7) and the off-grid net-zero condition in Eq. (9) using only PV and battery storage. The absence of biomass capacity in this case indicates that the reliability constraint in Eq. (18) is satisfied without dispatchable generation when storage capacity governed by Eqs. (13)–(16) is sufficient to manage load variability.

In contrast, the health center and agro-processing scenarios include biomass capacity. This outcome reflects the interaction between the reliability constraint in Eq. (18) and the cost trade-off embedded in Eq. (1). In these cases, the inclusion of dispatchable biomass generation reduces the required battery capacity while maintaining compliance with the LLP limit in Eq. (17). For the agro-processing plant, which exhibits sustained and productive energy demand, biomass capacity becomes economically favorable because it lowers storage oversizing that would otherwise be required to satisfy Eq. (7) during extended low-irradiance periods. The lower LCOE observed in larger systems reflects both scale effects in capital allocation under Eq. (2) and improved utilization of installed capacity.

4.2 Cost Structure and Binding Cost Components

The NPC decomposition indicates that capital investment, defined in Eq. (2), accounts for 65–78% of total lifecycle cost across all scenarios. This confirms that the objective function in Eq. (1) is primarily driven by upfront technology costs rather than operational expenditures. Battery systems contribute approximately 30–48% of the specified PV, battery, and biomass capital expenditure due to their dual role in satisfying hourly energy balance (Eq. (7)) and maintaining reliability (Eq. (18)). Replacement costs, represented in Eq. (3), are dominated by battery reinvestment over the 20-year project lifetime, whereas operation and maintenance costs in Eq. (4) remain comparatively modest for PV-dominated systems.

Fuel costs, represented in Eq. (5), influence only biomass-inclusive scenarios and remain secondary to capital cost contributions. Sensitivity testing confirms that, under baseline assumptions, capital cost parameters are binding

determinants of system configuration, whereas salvage value in Eq. (6) exerts negligible influence on optimal capacity selection.

4.3 Emission Reduction Outcomes and Constraint Activity

Table 3 presents the estimated emission intensity of the optimized system configurations relative to the diesel-only reference case. The results are evaluated using the emissions-accounting boundary described in Sections 2.6 and 3.7. Biogenic CO_2 released during conversion of sustainably sourced agricultural residues is excluded under the point-of-use carbon-neutrality assumption, whereas a simplified allowance for non-biogenic emissions from biomass collection, transport, handling, and preprocessing is included in the biomass-integrated case.

Table 3. Carbon dioxide emissions and reduction relative to diesel generation

System Type	CO_2 Emissions (kg/kWh)	Reduction Relative to Diesel
Diesel generator	0.82	–
PV + Battery	0.07	91%
PV + Biomass + Battery	0.09	89%

Note: PV = photovoltaic; "–" indicates not applicable.

The optimized renewable configurations achieve emission reductions of approximately 89–91% relative to diesel generation. Both values exceed the minimum 80% emission-reduction requirement imposed in the optimization. The emission constraint is therefore non-binding under the baseline assumptions, because the cost-minimizing renewable configurations already achieve substantially greater reductions than the minimum required level.

The PV–battery system achieves the lowest emission intensity at approximately 0.07 kg CO_2-e/kWh, corresponding to a reduction of approximately 91% relative to diesel generation. The biomass-integrated configuration has a slightly higher estimated intensity of approximately 0.09 kg CO_2-e/kWh, corresponding to an 89% reduction. This difference reflects the simplified inclusion of upstream biomass collection, transportation, handling, and preprocessing emissions rather than treating the entire biomass energy chain as completely emission-free. These results demonstrate that the optimized configurations can deliver substantial greenhouse gas reductions within the modeled system boundary. However, the values should not be interpreted as complete cradle-to-grave lifecycle assessments because equipment manufacturing, construction, land-use change, and other indirect effects are outside the present modeling boundary.

4.4 Sensitivity Analysis and Parameter Influence

Sensitivity analysis was conducted to evaluate the stability of the optimized solutions with respect to battery capital cost, financing conditions, and biomass fuel price (Table 4). Each parameter was varied independently while other baseline assumptions were maintained.

Table 4. Sensitivity of LCOE to key economic parameters

Parameter	Sensitivity Case	Approximate Effect on LCOE Relative to Baseline	Interpretation
Battery capital cost	−20%	−11% to −16%	Strong improvement in system economics
Battery capital cost	+20%	+10% to +14%	Significant deterioration in system economics
Real discount rate	6% instead of 8%	−6% to −9%	Lower financing cost improves viability
Real discount rate	12% instead of 8%	+9% to +14%	High financing cost significantly increases LCOE
Biomass fuel price	−15%	−3% to −5% in biomass cases	Moderate economic improvement
Biomass fuel price	+15%	+4% to +6% in biomass cases	Moderate cost increase

Note: LCOE = levelized cost of energy.

A 20% reduction in battery capital cost decreases LCOE by approximately 11–16% across the modeled configurations, confirming the strong influence of storage cost on both initial capital requirements and replacement

expenditure. Conversely, a 20% increase in battery capital cost raises LCOE by approximately 10–14%. The magnitude of this response is greatest in configurations with relatively high battery capacity, particularly the household and health-center cases. Financing conditions are also important. Reducing the real discount rate from the baseline value of 8% to 6% lowers LCOE by approximately 6–9%, whereas increasing the discount rate to 12% raises LCOE by approximately 9–14%. This result demonstrates that capital-intensive renewable systems remain strongly dependent on access to affordable long-term finance.

Biomass fuel-price uncertainty has a smaller influence than battery cost and financing conditions. A ±15% biomass fuel-price variation changes LCOE by approximately 3–6% in biomass-inclusive configurations. The relatively moderate response occurs because biomass fuel expenditure constitutes only one component of total lifecycle cost, whereas PV and battery capital expenditures account for a larger proportion of overall system cost. Despite these economic changes, the main technology-selection pattern remains stable over the tested sensitivity range. Solar PV remains the dominant generation technology, batteries remain essential for balancing variable renewable generation, and biomass remains most relevant in cases characterized by sustained institutional or productive-use demand. The reliability requirement also remains satisfied throughout the tested sensitivity cases.

4.5 Integrated Feasibility Assessment

Technical feasibility is demonstrated by the existence of feasible solutions that satisfy all operational constraints, including hourly energy balance (Eq. (7)), storage dynamics (Eqs. (13)–(16)), and reliability limits (Eq. (18)). No infeasible optimization outcomes were observed under baseline assumptions. Economic performance is assessed primarily through the reported levelized costs and lifecycle cost comparison. Discounted payback periods cannot be substantiated without an explicit avoided-cost or revenue baseline and annual cash-flow schedule; therefore, no payback estimates are reported. Environmental feasibility is confirmed by compliance with the emission reduction condition in Eq. (20), ensuring alignment with national decarbonization objectives. Notably, the emission constraint does not distort economic optimization, suggesting that low-carbon configurations are economically rational under current cost structures.

5 Discussion

This study provides a techno-economic assessment of decentralized net-zero energy systems under representative Ugandan conditions using an integrated MILP optimization framework. The findings demonstrate important interactions among solar photovoltaic generation, battery energy storage, dispatchable biomass generation, financing conditions, reliability requirements, and greenhouse gas mitigation. The discussion below interprets these findings in relation to existing literature, evaluates their robustness and policy significance, and identifies the methodological contributions and limitations of the study.

5.1 Dominance of Solar Photovoltaic and Consistency With Regional Studies

The optimization results demonstrate that solar photovoltaic generation forms the backbone of the cost-optimal net-zero energy configurations considered in this study. PV capacity is selected in all three representative demand cases, reflecting Uganda's favorable solar resource, the modularity of photovoltaic technology, and its comparatively low operating cost. This finding is consistent with previous studies of decentralized renewable-energy systems in Sub-Saharan Africa, where PV has frequently emerged as the dominant generation technology for residential, institutional, and rural electrification applications [3, 10, 12].

The estimated LCOE values of USD 0.072–0.118/kWh further support the economic competitiveness of PV-centered hybrid systems under the assumptions considered. Similar studies have shown that hybrid PV systems can provide economically attractive electricity in off-grid and weak-grid environments when technology sizing is optimized to local demand and resource conditions [10, 12]. The findings are also consistent with the broader decline in photovoltaic and battery-system costs identified in recent renewable-energy optimization studies [9].

A key contribution of the present study is that solar PV dominance is not imposed as an exogenous design assumption. Instead, it emerges endogenously from the MILP formulation after simultaneously accounting for lifecycle cost, hourly energy balance, reliability, annual net-zero requirements, and greenhouse gas constraints. This distinguishes the analysis from approaches in which technology combinations are predefined before simulation. The result therefore provides additional evidence that PV is structurally attractive within the representative Ugandan applications considered, rather than merely being favored because of an assumed system architecture. Nevertheless, this conclusion should not be interpreted as implying that PV-dominated configurations will necessarily be optimal at every location in Uganda. Local electricity demand, solar resource variability, grid conditions, financing arrangements, land availability, and access to alternative renewable resources may change the relative attractiveness of different technologies.

5.2 Role of Battery Storage and Alignment With Cost-Sensitivity Literature

Battery energy storage is a critical enabling technology in the optimized net-zero systems, particularly in the household and institutional cases. Storage is required to manage the temporal mismatch between daytime solar electricity generation and demand occurring during evening, nighttime, and low-irradiance periods. This finding is consistent with the growing literature identifying energy storage as a central component of high-renewable electricity systems and, simultaneously, as one of their most important cost constraints [9, 11].

In the baseline configurations, battery systems account for a substantial proportion of initial capital expenditure. The quantitative sensitivity analysis reinforces their economic significance. A 20% reduction in battery capital cost decreases LCOE by approximately 11–16%, whereas a 20% increase in battery cost increases LCOE by approximately 10–14%. These responses demonstrate that relatively modest changes in storage prices can materially influence overall system economics. This behavior is consistent with other techno-economic studies showing that battery capital and replacement costs strongly affect the economic performance of renewable hybrid microgrids [9, 11].

The results also demonstrate that battery requirements depend strongly on the availability of dispatchable generation. In the agro-processing case, biomass generation complements PV production during low-solar periods and reduces the extent to which additional battery capacity is required solely for reliability. This interaction is important because many PV-centered hybrid-system studies focus predominantly on increasing storage capacity to compensate for renewable intermittency. The present results indicate that, in productive-use applications with sufficiently high and sustained demand, dispatchable renewable generation can provide an economically meaningful alternative to excessive battery oversizing.

The sensitivity results therefore have practical implications beyond technology selection. Continued declines in lithium-ion battery costs would substantially improve the competitiveness of decentralized net-zero systems, while unexpectedly high battery prices could delay economic deployment. Improvements in battery lifetime, recycling, replacement cost, and supply-chain availability could consequently have a significant influence on the long-term economics of decentralized renewable systems in Uganda.

5.3 Biomass Integration: Agreement and Contrast With Existing Literature

The selection of biomass generation in the institutional and agro-processing configurations is broadly consistent with studies emphasizing the role of dispatchable bioenergy in improving reliability and reducing dependence on storage in renewable-energy systems [8, 28, 29]. Uganda has substantial agricultural activity, creating potential opportunities for energy recovery from residues where feedstock availability, collection systems, and conversion technologies are technically and economically suitable. The role of biomass is particularly relevant in the agro-processing case because productive-use loads tend to be larger and more sustained than residential demand. Biomass generation can supply controllable electricity during periods when solar generation is insufficient, thereby reducing the storage capacity that would otherwise be required to maintain the reliability constraint. This finding is consistent with research highlighting synergies between agricultural residues, decentralized energy production, and productive rural energy uses [14, 30, 31].

However, the results also indicate that biomass should not be regarded as universally cost-optimal. The household configuration does not select biomass, demonstrating that the additional capital cost, fuel requirements, and operational complexity of dispatchable bioenergy are not justified under relatively low demand. Biomass becomes more attractive where electricity demand is sufficiently large or continuous to make regular utilization of the generation unit economically worthwhile. This result contrasts with generalized approaches that treat biomass as a default baseload resource for rural mini grids.

An important consideration is the biomass supply chain. Agricultural residues must generally be collected, aggregated, transported, stored, and in some cases dried or otherwise preprocessed before conversion. Biomass supply-chain studies have demonstrated that logistics and preprocessing arrangements can materially influence the economic and environmental performance of biomass energy systems [17, 22]. Consequently, proximity between residue sources and the conversion facility, feedstock density, transport distance, seasonal availability, and competing uses of residues may determine whether biomass integration remains advantageous in practice.

The environmental accounting used in the present study therefore distinguishes between biogenic carbon released during conversion and upstream emissions associated with fuel provision. Combustion of sustainably sourced agricultural residues is treated as carbon-neutral at the point of electricity generation, while a simplified allowance is made for non-biogenic emissions associated with collection, transport, handling, and preprocessing. This treatment recognizes that carbon neutrality at the point of use does not imply that the complete biomass supply chain is emission-free. Accordingly, the study supports context-specific, rather than universal, use of biomass. Biomass appears particularly valuable where agricultural residues are sustainably available near the point of use and where sufficiently continuous electricity demand can justify dispatchable generation. Site-specific feedstock and logistics assessment would therefore be necessary before project-level deployment.

5.4 Emission Reduction Outcomes and Implications for Climate Policy

The optimized renewable-energy configurations achieve greenhouse gas emission reductions of approximately 89–91% relative to diesel-based electricity generation, as reported consistently in Table 3. Specifically, the PV–battery configuration achieves an estimated reduction of approximately 91%, while the PV–biomass–battery configuration achieves approximately 89%. Both configurations therefore exceed the minimum 80% emission-reduction requirement imposed in the optimization framework. The substantial reduction relative to diesel generation is consistent with studies showing that renewable hybrid energy systems can provide significant environmental benefits in developing-country applications where diesel generation would otherwise serve as the principal or backup electricity source [4]. More broadly, the results are aligned with analyses emphasizing the potential for low-carbon electricity technologies to contribute to decarbonization pathways across African energy systems [13].

The slightly higher emission intensity of the biomass-integrated configuration requires careful interpretation. Biomass combustion is treated as carbon-neutral at the point of use when sustainably sourced agricultural residues are utilized. However, the biomass supply chain can generate fossil greenhouse gas emissions through collection, transportation, handling, drying, and preprocessing. The emissions estimate for the biomass-integrated configuration therefore includes a simplified allowance for these upstream activities rather than assuming that all biomass-related processes have zero emissions. This treatment is consistent with the broader recognition that biomass sustainability depends not only on combustion characteristics but also on feedstock sourcing and supply-chain organization [17, 22].

An important result is that the 80% emission-reduction constraint is non-binding under the baseline assumptions. The cost-minimizing solutions independently achieve emission reductions substantially greater than the required threshold. This suggests that, within the representative cases examined, economic optimization and environmental performance are strongly aligned. Similar arguments have been made in the decentralized-energy literature, where declining renewable-energy costs increasingly allow electrification and climate-mitigation objectives to be pursued simultaneously [3, 12].

These results have potential relevance to Uganda's broader decarbonization objectives because replacing diesel-based electricity with renewable hybrid systems could support both energy-access and emissions-reduction goals. However, the estimated 89–91% reduction range should be interpreted within the defined system boundary. The analysis is not a complete cradle-to-grave lifecycle assessment and does not comprehensively account for emissions associated with manufacturing of PV modules and batteries, construction, land-use effects, detailed component end-of-life processes, or location-specific biomass logistics. Consequently, the results demonstrate the emissions-reduction potential of the representative systems rather than a universal lifecycle emission factor applicable to all projects in Uganda.

5.5 Financing Sensitivity and Institutional Implications

The sensitivity analysis demonstrates that financing conditions are another major determinant of net-zero system economics. Increasing the real discount rate from the baseline value of 8% to 12% raises LCOE by approximately 9–14%, whereas reducing the discount rate to 6% lowers LCOE by approximately 6–9%. The magnitude of these changes reflects the capital-intensive nature of PV and battery systems, in which a large proportion of total expenditure occurs at the beginning of the project. This finding is consistent with the broader techno-economic literature showing that renewable hybrid systems can be highly sensitive to financing assumptions because capital costs are incurred upfront while economic benefits accrue over many years [9]. The influence of financing is particularly important in developing-country contexts, where borrowing costs, risk premiums, currency uncertainty, and access to long-term capital can differ substantially from those prevailing in mature renewable-energy markets.

The results indicate that technology-cost reductions alone may therefore be insufficient to achieve the full economic potential of decentralized net-zero systems. Lower-cost financing can substantially reduce annualized system costs even without changes in technical performance. Conversely, high discount rates can erode the apparent cost advantage of technologies characterized by low operating expenditure but relatively high initial capital requirements. From an institutional perspective, these findings support consideration of concessional finance, credit guarantees, blended-finance mechanisms, green credit facilities, and other instruments aimed at lowering project financing costs. Such mechanisms could be particularly valuable for battery-intensive systems, where both the technology cost and the cost of capital exert strong influences on LCOE. However, the specific effectiveness of individual financing interventions would depend on project scale, investor risk perception, tariff structure, currency exposure, and institutional arrangements and therefore requires project-specific evaluation.

5.6 Methodological Advances and Unique Contributions

Beyond the numerical findings, the study contributes methodologically to the literature on decentralized renewable and net-zero energy-system planning.

First, it applies an integrated MILP framework that simultaneously determines technology capacity and operating decisions while enforcing annual net-zero energy balance, hourly supply-demand equilibrium, reliability

requirements, and greenhouse gas emission constraints. MILP approaches are increasingly used for complex energy-system planning because they allow technical, economic, and operational decisions to be represented within a common optimization framework [16, 20].

Second, the study compares three distinct electricity-use contexts within one consistent framework: a household case, an institutional health-center case, and an agro-processing case. Previous studies in the region have often examined applications independently, including rural hybrid electrification, health-facility PV systems, and biomass-energy utilization [3, 10, 32]. By applying the same optimization logic across multiple demand categories, the present analysis allows the influence of load characteristics on technology selection to be assessed more directly.

Third, environmental performance is integrated into the optimization through an explicit minimum emission-reduction constraint rather than being evaluated solely as an ex-post indicator. This allows the model to test whether achieving a defined emissions target alters the cost-optimal system configuration. The result that the emission constraint is non-binding under baseline conditions is itself informative, because it indicates that the economic optimum already satisfies the specified environmental threshold.

Fourth, the framework explicitly links the observed technology configurations to the mathematical constraints responsible for their selection. Battery storage is connected to hourly balancing and reliability requirements, biomass to dispatchability and fuel-cost considerations, and grid exchange to the annual net-zero condition. This improves transparency compared with optimization studies where model outputs are reported without clearly identifying the constraints that drive system choices. Finally, the framework provides a basis for subsequent extension to more complex formulations. Future studies could incorporate stochastic weather and demand, multiple representative years, endogenous demand growth, spatial biomass logistics, unit-commitment constraints, dynamic tariffs, and probabilistic reliability. Such developments would further strengthen the applicability of optimization-based approaches for decentralized energy planning in Uganda and comparable Sub-Saharan African settings.

5.7 Limitations and Directions for Future Research

While the results are robust within the defined modeling framework, several limitations warrant consideration.

First, the analysis uses hypothetical representative load profiles rather than measured high-resolution demand data from individual households, health facilities, or agro-processing plants. The profiles reproduce typical operating characteristics but may not capture all site-specific behavioral variations, equipment start-up loads, demand growth, or unusual operating schedules.

Second, deterministic typical meteorological data are used to represent renewable resource conditions. The absence of stochastic weather modeling may understate the effects of prolonged low-irradiance events, extreme weather conditions, and interannual climate variability on system reliability and storage requirements.

Third, the geographical analysis is based on three representative cases—Kampala, Gulu District, and Mbarara District—selected to characterize grid-connected, rural/off-grid, and agro-processing conditions. These cases do not capture the complete diversity of electricity infrastructure, renewable resources, income levels, demand patterns, biomass availability, and financing conditions across Uganda.

Fourth, biomass combustion is treated as carbon-neutral at the point of use for sustainably sourced agricultural residues, while only simplified upstream emissions associated with collection, transportation, handling, and preprocessing are represented. A complete lifecycle assessment of biomass supply, including location-specific transport distances, soil-carbon effects, competing residue uses, land-use implications, and detailed processing energy requirements, is outside the scope of the present study.

Fifth, the modeling framework assumes relatively stable technology performance, financing conditions, biomass availability, and regulatory arrangements over the study horizon. Future changes in tariffs, net-metering rules, battery prices, grid emission intensity, interest rates, equipment availability, and fuel markets could alter the optimal configurations.

Future research should therefore incorporate measured site-level demand profiles, probabilistic renewable-resource modeling, demand-growth trajectories, spatially explicit biomass supply-chain assessment, alternative financing structures, and dynamic policy scenarios. Such extensions would increase the applicability of the framework to project-level investment assessment.

6 Policy Implications

The findings of this study provide evidence for planning decentralized low-carbon electricity systems under the representative Ugandan conditions examined. The results should be interpreted as indications of technology and policy priorities rather than as evidence that a single system configuration is universally optimal throughout Uganda. Across the modeled residential, institutional, and productive-use cases, solar photovoltaic generation consistently forms the principal renewable generation technology. This result suggests that policies aimed at reducing the cost and administrative burden of PV deployment can support wider adoption of decentralized electricity systems. Measures may include efficient customs procedures, appropriate fiscal incentives, quality standards, improved

installer certification, and stronger technical support for solar and storage systems. Battery storage is also central to the modeled systems, particularly where solar PV must serve demand outside daylight hours. The sensitivity analysis shows that a 20% reduction in battery capital cost decreases LCOE by approximately 11–16%, whereas a 20% increase raises LCOE by approximately 10–14%. Policies that encourage technology cost reduction, competitive procurement, battery-quality assurance, recycling infrastructure, and access to appropriate financing could therefore have a substantial effect on system affordability.

Financing conditions are similarly important. Raising the real discount rate from 8% to 12% increases LCOE by approximately 9–14%, demonstrating the sensitivity of capital-intensive renewable systems to the cost of finance. Concessional lending, blended finance, credit guarantees, green investment facilities, and other risk-reduction mechanisms could improve project economics where commercial borrowing costs remain high. The results also suggest that biomass should be promoted selectively rather than uniformly. Biomass integration is most attractive in applications with sustained electricity demand and adequate access to agricultural residues, such as selected agro-processing activities. Policy support should therefore prioritize sustainable residue assessment, aggregation systems, efficient transport and preprocessing, quality standards, and safeguards against unsustainable extraction or competition with existing agricultural uses.

The modeled biomass results additionally demonstrate the importance of accounting for the full supply chain. Although combustion of sustainably sourced residues is treated as carbon-neutral at the point of generation, residue collection, transportation, handling, and preprocessing can create greenhouse gas emissions and additional costs. Biomass-support programs should therefore consider both feedstock sustainability and supply-chain efficiency when assessing potential projects. The finding that the optimized configurations achieve emission reductions of approximately 89–91% relative to diesel generation indicates that substantial decarbonization can accompany decentralized electricity provision under the modeled assumptions. However, this result should not be interpreted as a guaranteed emission-reduction range for all Ugandan projects. Actual performance will depend on local renewable resources, demand patterns, technology choices, biomass logistics, grid characteristics, financing conditions, and operating practices.

The policy value of the present modeling framework lies primarily in technology screening, comparative scenario analysis, and identification of the parameters that most strongly influence system economics and environmental performance. National and local energy agencies could use similar optimization approaches to compare electrification strategies, identify promising technology combinations, and prioritize locations for more detailed feasibility assessment. Accordingly, the findings support planning-level decisions rather than direct investment decisions for individual sites. Before project implementation, developers and public agencies should undertake site-specific electricity-demand measurements, renewable-resource assessment, network and infrastructure evaluation, biomass supply-chain analysis where relevant, detailed financial modeling, environmental assessment, and stochastic reliability analysis.

7 Conclusion

This study developed and applied a MILP framework to optimize decentralized net-zero energy systems under representative Ugandan climatic, economic, and operational conditions. The framework integrates lifecycle cost minimization, annual net-zero energy balance, hourly supply-demand balancing, reliability requirements, and greenhouse gas emission constraints within a unified optimization structure. Three representative settings were examined: Kampala in the Central Region to characterize a predominantly grid-connected context, Gulu District in the Northern Region to represent rural and off-grid electrification conditions, and Mbarara District in the Western Region to represent agro-processing and productive-use electricity demand. These locations were selected to provide contrasting planning cases and should not be interpreted as representing every site within their respective regions.

The results demonstrate that solar photovoltaic generation forms the principal generation technology in all modeled configurations. Battery energy storage provides an essential balancing and reliability function, particularly for household and institutional applications. Biomass generation becomes economically attractive in scenarios with higher and more sustained electricity demand, where dispatchable generation can reduce dependence on battery oversizing and provide additional operational flexibility. The optimized configurations produce LCOE values ranging from USD 0.072 to 0.118/kWh. The sensitivity analysis demonstrates that battery cost and financing conditions are the most influential economic parameters. A 20% reduction in battery capital cost reduces LCOE by approximately 11–16%, whereas a 20% increase raises LCOE by approximately 10–14%. Increasing the real discount rate from 8% to 12% increases LCOE by approximately 9–14%, confirming the importance of affordable financing for capital-intensive renewable-energy systems. Biomass fuel-price variations have a smaller effect on overall system economics.

Environmental performance is also favorable within the defined system boundary. The optimized PV–battery and PV–biomass–battery configurations achieve approximately 91% and 89% emission reductions, respectively, relative to diesel generation. Both values exceed the minimum 80% reduction requirement incorporated into the optimization.

Biomass combustion is treated as carbon-neutral at the point of use for sustainably sourced agricultural residues, while simplified emissions associated with residue collection, transportation, handling, and preprocessing are recognized in the biomass-integrated case. These findings demonstrate the techno-economic potential of decentralized net-zero energy systems within the representative Ugandan cases evaluated. They do not establish that identical configurations or economic outcomes will apply across all locations in Uganda. The use of representative sites, hypothetical load profiles, deterministic weather data, simplified biomass lifecycle accounting, and fixed economic assumptions limits direct extrapolation to individual projects.

The principal contribution of the study is therefore to provide a transparent optimization framework for technology screening, energy planning, and comparative scenario assessment. The results can assist policymakers, planners, researchers, and project developers in identifying promising combinations of solar PV, battery storage, and context-specific biomass generation and in understanding the technological and financial factors that most strongly influence system performance. Before applying the results to an individual investment decision, site-specific analysis should be undertaken using measured electricity demand, local renewable-resource data, actual financing conditions, infrastructure constraints, biomass supply-chain characteristics where applicable, and probabilistic assessment of operational uncertainty. Future research incorporating these factors would further strengthen the use of MILP-based optimization for detailed decentralized-energy project design in Uganda.

Data Availability

Not applicable.

Conflicts of Interest

The author declares no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of the manuscript, the authors used ChatGPT, a large language model developed by OpenAI, to assist with language editing, grammatical correction, readability, and organization of author text. All outputs were critically reviewed, verified against the original data and references, and revised by the authors, who take full responsibility for the final content.

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Nomenclature

Symbol	Description	Unit
t	Hourly time index	h
Δt	One hour	h
T	Total simulation hours (8760)	h
y	Year index	year
N	Project lifetime	years
r	Real discount rate	–
X_i	Installed capacity of component i	kW or kWh
L_t	Electrical load demand at time t	kWh
P_t^{PV}	PV power output	kW
P_t^{Bio}	Biomass generator output	kW
P_t^{Wind}	Wind turbine output	kW
P_t^{unmet}	Unmet demand	kW
P_t^{loss}	Battery conversion and other modeled power losses	kW
P_t^{curt}	Curtailed renewable power	kW
$P_t^{\mathrm{Bat,ch}}$	Battery charging power	kW
$P_t^{\mathrm{Bat,dis}}$	Battery discharging power	kW
$P_t^{\mathrm{Grid,imp}}$	Grid electricity import	kW
$P_t^{\mathrm{Grid,exp}}$	Grid electricity export	kW
SOC_t	Battery state of charge	kWh
$\Delta\mathrm{SOC}$	Net change in stored energy	kWh
η_{PV}	PV conversion efficiency	–
$\eta_{\mathrm{ch}}, \eta_{\mathrm{dis}}$	Battery charge/discharge efficiency	–
η_{Bio}	Biomass electrical efficiency	–
G_t	Plane-of-array solar irradiance	W/m^2
G_{STC}	Irradiance at standard test conditions	W/m^2
F_t	Fuel consumption	kg or L
EF	Emission factor	kg CO_2/unit
NPC	Net present cost	USD
LCOE	Levelized cost of energy	USD/kWh
LLP	Loss of load probability	–

Note: "–" indicates not applicable.